

⇒ Rocket Traj. Sim. Exercise 9:

Starting from the previous exercise, replace the initial offset from vertical with a vertical lift-off followed by a pitch-over (i.e. implement a "pitch program").

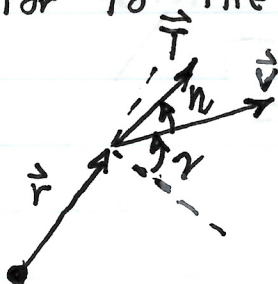
> Flight Path Angle & Steering Angle

- Flight path angle: angle measured from local horizon to the velocity vector

$$\gamma = \frac{\pi}{2} - \arccos\left(\frac{\vec{r} \cdot \vec{v}}{\|\vec{r}\| \|\vec{v}\|}\right) = \arcsin\left(\frac{\vec{r} \cdot \vec{v}}{\|\vec{r}\| \|\vec{v}\|}\right)$$



- Steering angle: angle measured from the velocity vector to the thrust vector, represented as η



for 2D traj. sim. initial pitchover...

if launch azimuth = 90° , then $\eta > 0$

if " " " = -90° , then $\eta < 0$

- Applying η to velocity norm vector used in calculating thrust direction:

$$R_z(\eta) = \begin{bmatrix} \cos(\eta) & -\sin(\eta) \\ \sin(\eta) & \cos(\eta) \end{bmatrix}$$

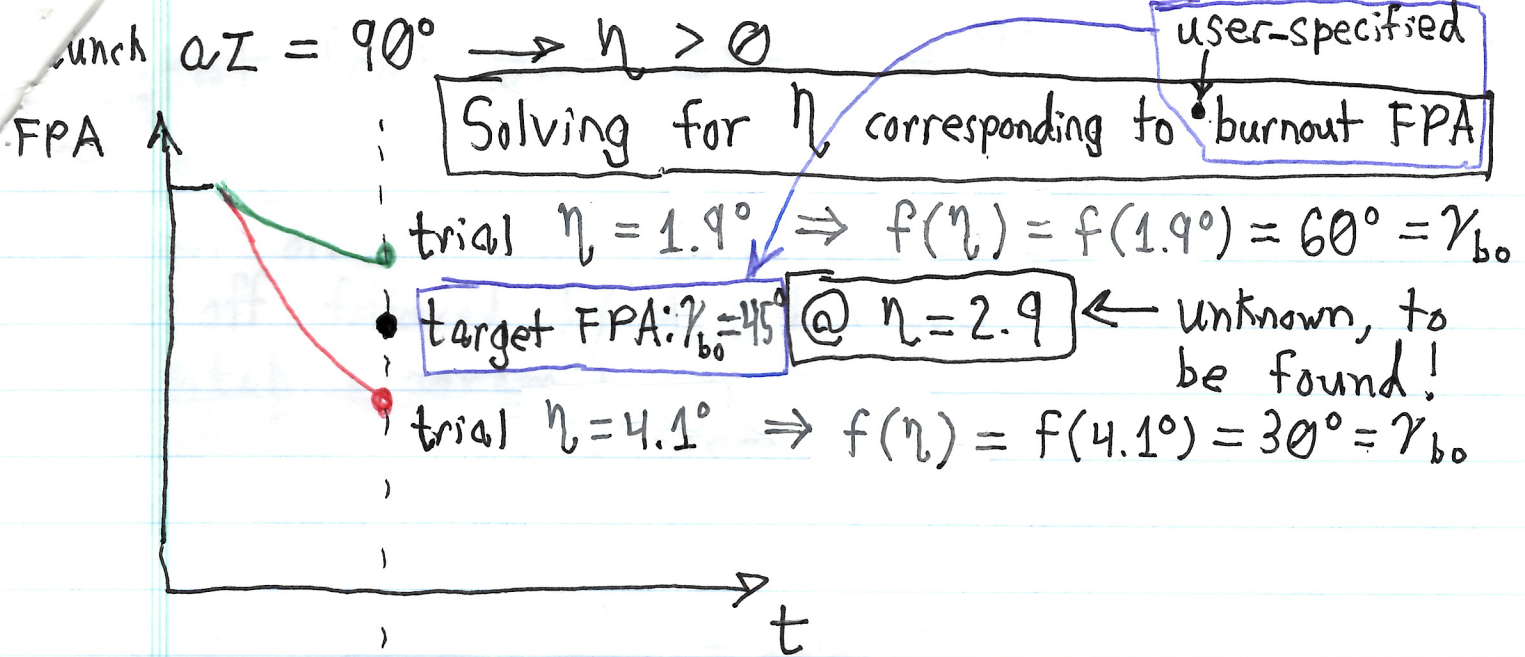
$$\hat{\vec{v}} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{[v_x, v_y]}{\sqrt{v_x^2 + v_y^2}}$$

$$\hat{\vec{v}}_\eta = R_z \hat{\vec{v}}$$

$$= \begin{bmatrix} v_x \cos(\eta) - v_y \sin(\eta) \\ v_x \sin(\eta) + v_y \cos(\eta) \end{bmatrix} / \sqrt{v_x^2 + v_y^2} = \begin{bmatrix} v_{\eta,x} \\ v_{\eta,y} \end{bmatrix}$$

Thus, acceleration due to thrust is given by:

$$\vec{a}_{\text{thrust}} = a_{\text{thrust},x} v_{\eta,x} \hat{i} + a_{\text{thrust},y} v_{\eta,y} \hat{j}$$



~~$\eta_{guess} = 4.1$~~

$$\begin{aligned} \text{FPA diff} &= \text{trial FPA} - \text{target FPA} \\ &= 30 - 45 \\ &= -15 \end{aligned}$$

$$\begin{aligned} \eta_{guess} &= + \text{FPA diff} \times \frac{\eta_{guess} - \eta_{target}}{15} + \eta_{guess} \\ &= + \text{FPA diff} \times \frac{1.2}{15} + \eta_{guess} \end{aligned}$$

burnout flight path angle solved for a given η in propagator function

FPA = $f(\eta)$

⇒ Secant method for root finding:

$$\eta_{n+1} = \eta_n - \frac{f(\eta_n)}{f'(\eta_n)}$$

$$f'(\eta_n) = \frac{f(\eta_n) - f(\eta_{n-1})}{\eta_n - \eta_{n-1}}$$

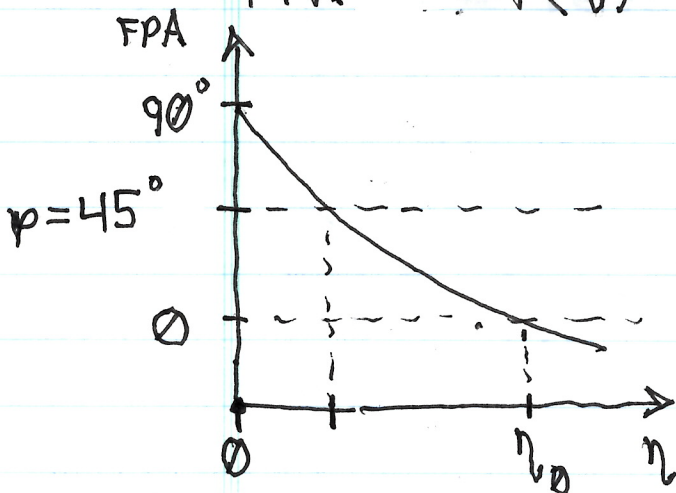
$$\eta_{n+1} = \eta_n - \frac{f(\eta_n)(\eta_n - \eta_{n-1})}{f(\eta_n) - f(\eta_{n-1})}$$

⇒ Modified secant method

$$\eta_{n+1} = \eta_n + \frac{(p - f(\eta_n))(\eta_n - \eta_{n-1})}{f(\eta_n) - f(\eta_{n-1})}$$

Where

p is target flight path angle



use this!